

Patient Shortage in Histopathology: Does Cohort Composition Matter at a Fixed Patient Count?

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Abstract

TCGA-UT patch classifiers trained on more patients are more accurate than redundancy-adjusted effective support predicts, and tissue source sites explain little of this residual. This experiment asks whether the composition of a patient cohort matters when its size is fixed. Three cohorts of twenty patients with eight patches each share one randomly drawn focal patient. The remaining patients are its nearest neighbours in the patient-mean Virchow2 embedding (*clustered*), randomly drawn patients (*random*), or patients selected to cover the class’s training patients as evenly as possible (*dispersed*). Patients, patches, and effective support are equal across the cohorts. The clustered cohort reaches 62.99 percent patient-macro balanced accuracy, 8.30 percentage points below the random cohort (test-patient 95% interval: 7.66 to 8.95), nearly the full advantage of twenty over five patients at the same budget. This damage falls on test patients far from the clustered cohort. The dispersed cohort improves on the random cohort by 1.04 points (0.44 to 1.66), an interval that crosses the one-point practical threshold. Cohort composition therefore matters at a fixed patient count, mainly as protection against a poorly covering cohort, although redundancy between similar patients may contribute to the size of the damage.

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1 Motivation

Spreading a labelling budget over more patients improves TCGA-UT patch classification beyond what redundancy explains. At 160 patches per class, twenty patients contributing eight patches each outperform five patients contributing 32 patches each by 9.52 percentage points of patient-macro balanced accuracy (Queisler, 2026b). At equal effective support, each doubling of patients still adds 2.62 points (Queisler, 2026c), about 5.2 points over the two doublings of this comparison. Doubling tissue source sites at a fixed patient count adds 1.06 points, mostly for test patients from the added sites, and twenty patients from five sites are at least as accurate as twenty randomly drawn patients (Queisler, 2026e).

Patient coverage, the share of a cancer type’s between-patient variation that the training patients represent, remains the leading label-free candidate. Two preceding experiments built twenty-patient cohorts around the same five patients and added either patients similar to them or randomly drawn patients (Queisler, 2026d,a). The similar additions covered held-out patients almost as well as the random ones: they attained 84 to 88 percent of the coverage improvement of random additions, and still 67 to 73 percent when all additions resembled a single patient. Quadrupling the patient count shortens the distance from most held-out patients to their nearest training patient, whichever patients are added, so with this coverage measure patient count and coverage cannot be varied independently.

Coverage does, however, vary strongly between cohorts of equal size. Twenty patients forming one neighbourhood cover validation patients at a mean distance of 0.346 to 0.356, twenty random patients at 0.292 to 0.296, and twenty patients selected to cover the training pool as evenly as possible at 0.268 to 0.271; five random patients cover them at 0.385 to 0.387 (Queisler, 2026a). This report therefore replaces the question whether similar patients help beyond effective support with a question that the measure can address: *at a fixed patient count, does a cohort that covers the class’s patients poorly train a worse classifier, and does one that covers them well train a better classifier than random sampling?*

2 Study design

Only the composition of a twenty-patient cohort changes. The design inherits from Queisler (2026a) the 30 eligible TCGA-UT cancer types, the frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), the three fixed patient-disjoint splits with their natural validation and test partitions, the nested round-robin patch order, and the readout: multinomial logistic regression with its regularization strength selected on the validation partition for every classifier. It also inherits the patient distance. For split r and class c , $\bar{\mathbf{x}}_{irc}$ is the mean feature vector of patient i ’s patches of the class, $\boldsymbol{\mu}_r$ the average of these means over all classes, and

$$\mathbf{e}_{irc} = \frac{\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r}{\|\bar{\mathbf{x}}_{irc} - \boldsymbol{\mu}_r\|}, \quad d(i, j) = 1 - \mathbf{e}_{irc}^\top \mathbf{e}_{jrc}. \quad (1)$$

Allocation	G	m	Patients
Clustered	20	8	focal patient, its 19 nearest neighbours
Random	20	8	focal patient, 4 other anchors, 15 random patients
Dispersed	20	8	focal patient, 19 coverage-maximizing patients

Table 1: The three allocations share the focal patient, the patient count G , the patches per patient m , the budget of 160 patches per class, and therefore the effective support. They differ only in how the remaining nineteen patients are chosen, which moves coverage from low to high.

Table 1 lists the allocations. Because all three share $G = 20$ and $m = 8$, effective support is identical by construction, and no effective-support surface enters the analysis.

2.1 Allocation rules

Three rules define the training sets and check the manipulation before any classifier is trained. First, every class is admitted. The pool $\mathcal{E}_{rc}$ contains the training patients with at least 32 patches of class c ; pools hold between 30 and 514 patients, so every class can supply twenty.

Second, the allocations share a draw structure. For each split, class, and draw $d \in \{1, \dots, 5\}$, the five *anchor patients* are the patients of the stored five-patient draw of Queisler (2026b). The *focal patient* f is one of these five anchors, drawn uniformly at random as in Queisler (2026a), and it is the only patient that every allocation contains by construction. The random allocation adds fifteen randomly drawn pool patients to the five anchors, as in Queisler (2026a), so it contains f and is distributed like a twenty-patient random draw. The clustered allocation contains f and the nineteen patients of $\mathcal{E}_{rc} \setminus \{f\}$ closest to it under Equation (1). The dispersed allocation starts from $S_0 = \{f\}$ and adds nineteen patients greedily by *coverage-maximizing selection*, known as facility location in operations research:

$$S_{k+1} = S_k \cup \left\{ \arg \min_{j \in \mathcal{E}_{rc} \setminus S_k} \frac{1}{|\mathcal{E}_{rc}|} \sum_{i \in \mathcal{E}_{rc}} \min_{l \in S_k \cup \{j\}} d(i, l) \right\}. \quad (2)$$

Each step adds the pool patient that most shortens the mean distance from pool patients to their nearest selected patient. The rule uses only distances in the embedding, not tissue source sites. Ties in both rules are broken by patient identifier, and neither rule uses validation or test patients. Every patient contributes the first eight patches of its round-robin order.

Third, a manipulation check compares how well the allocations cover held-out patients. The *coverage distance* of a patient set is the mean distance from the class’s validation patients $\mathcal{V}_{rc}$ to their nearest training patient,

$$r_{rc}(\mathcal{P}) = \frac{1}{|\mathcal{V}_{rc}|} \sum_{i \in \mathcal{V}_{rc}} \min_{j \in \mathcal{P}} d(i, j). \quad (3)$$

Let r_{clu} , r_{ran} , and r_{dis} be the coverage distances of the three allocations and r_5 that of the five anchors, each averaged over classes and draws within a split. The difference $r_5 - r_{ran}$ is the coverage gained when fifteen random patients join the five anchors, and it serves as the yardstick for both contrasts. The two *coverage ratios*

$$\rho_{con} = \frac{r_{clu} - r_{ran}}{r_5 - r_{ran}}, \quad \rho_{sel} = \frac{r_{ran} - r_{dis}}{r_5 - r_{ran}} \quad (4)$$

express the coverage change behind each contrast as a fraction of this gain. Fitting proceeds only if $\rho_{con} \geq 0.5$ in all three splits. The ratio ρ_{sel} , the class-level ratios, the mean pairwise distance within each cohort, and the share of patients from the focal patient’s tissue source site are recorded without a threshold.

Table 2 reports these quantities as computed by Queisler (2026a) from the same embeddings, draws, and selection rules, before any classifier of this experiment was trained. The check is expected to pass: ρ_{con} lies between 0.58 and 0.68, and the class-level ratio reaches 0.5 in 52 of 90 class-split combinations. The secondary contrast rests on a much smaller manipulation, about a quarter of the coverage gained from five to twenty random patients, which constrains its interpretation in Section 4.

3 Evaluation

Both contrasts are read with the same accuracy measure and the same uncertainty procedure.

Split	Coverage distance					
	r_5	r_{clu}	r_{ran}	r_{dis}	ρ_{con}	ρ_{sel}
1	0.387	0.353	0.296	0.271	0.63	0.27
2	0.385	0.356	0.293	0.268	0.68	0.27
3	0.385	0.346	0.292	0.268	0.58	0.26

Table 2: Clustering moves coverage far more than coverage-maximizing selection does. Coverage distances are mean cosine distances from validation patients to their nearest training patient, averaged over the 30 classes and five draws, as computed by Queisler (2026a). Mean pairwise distances within a cohort are 0.35 to 0.37 for clustered, 0.59 to 0.60 for random, and 0.67 for dispersed allocations.

3.1 Concentration damage and selection gain

Accuracy is patient-macro balanced accuracy over all 30 classes: patch recall is averaged within each test patient and true class, then across patients within each class, and finally across classes. Averaging three splits and five draws gives A_{clu} , A_{ran} , and A_{dis} in percent. The primary endpoint is the *concentration damage*

$$\Delta_{\text{con}} = A_{\text{ran}} - A_{\text{clu}}, \quad (5)$$

the accuracy lost when nineteen patients resemble one patient instead of being drawn at random. The secondary endpoint is the *selection gain*

$$\Delta_{\text{sel}} = A_{\text{dis}} - A_{\text{ran}}, \quad (6)$$

the accuracy won when coverage-maximizing selection replaces random sampling. Both hold patients, patches per patient, budget, and effective support fixed.

A rough scale helps to read Δ_{con} . If coverage alone carried the residual of about 5.2 points between five and twenty random patients, and accuracy changed linearly with coverage distance, the concentration damage would be about $\rho_{\text{con}} \times 5.2 \approx 3$ points. This scale is descriptive, assumes a linearity that no preceding experiment tested, and carries no decision threshold.

Two secondary analyses describe where a concentration damage arises. First, each test patient receives per draw its *coverage improvement*, the distance to its nearest clustered training patient minus the distance to its nearest random training patient, and Δ_{con} is computed within tertiles of this improvement with the patient weights of the primary endpoint. Patient coverage predicts a damage that grows with the improvement. Second, all quantities, including the class-level coverage ratios, are reported separately for the thirteen site classes of Queisler (2026e) and the seventeen remaining classes. These analyses carry no decision threshold.

3.2 Uncertainty

Uncertainty follows Queisler (2026b). Each of 2,000 paired Bayesian bootstrap replicates assigns Dirichlet(1, ..., 1) weights to the distinct test patients (Rubin, 1981), shared across allocations, splits, draws, and classes. The 2.5th and 97.5th percentiles give exploratory 95% intervals. Embeddings and allocations are held fixed. The standard deviation of each allocation’s accuracy and of both contrasts across the fifteen split–draw combinations is reported as a descriptive measure of sensitivity to the drawn patients.

4 Interpretation criteria

Conclusions rest on the test-patient intervals for Δ_{con} and Δ_{sel} and on the practical threshold of one percentage point used in the preceding experiments (Table 3). The two contrasts are

not treated symmetrically. Concentration damage rests on a coverage change of more than half the coverage gained from five to twenty random patients, so an interval within $[-1, 1]$ counts as practical equivalence. Selection gain rests on about a quarter of that gain, so only an interval above one point is informative; an interval within $[-1, 1]$ does not show that coverage-guided selection is useless.

Reading	Claim	Concentration damage Δ_{con}	Selection gain Δ_{sel}
Selection beyond sampling	Composition matters, and coverage-guided selection improves on random sampling	Interval above 1	Interval above 1
Concentration damage	Composition matters, but only a poorly covering cohort is shown to hurt	Interval above 1	Interval not above 1
No composition effect	Patient-mean coverage does not affect accuracy at twenty patients	Interval within $[-1, 1]$	Interval not above 1

Table 3: Each reading predicts a distinct combination of the two contrasts. Intervals are test-patient 95% intervals in percentage points. All other combinations, including a concentration-damage interval that crosses a threshold, are inconclusive.

Selection beyond sampling. If both contrasts exceed the threshold, patient-mean coverage separates useful from less useful cohorts in both directions from random sampling. Coverage distance would then be a candidate shortage signal alongside effective support, and the next step would compare coverage-selected and random cohorts of five patients with 32 patches each, the setting in which patient shortage was first observed.

Concentration damage. If only the concentration damage exceeds the threshold, a cohort that misses most of the class’s between-patient variation trains a worse classifier, but random sampling may already capture most of the attainable coverage at twenty patients. Coverage would remain a plausible explanation of the breadth benefit; its value for patient selection would still be open and would again be tested first at five patients, where random cohorts cover the class less well.

No composition effect. If the concentration damage is practically zero, a coverage change of more than half the coverage gained from five to twenty random patients leaves accuracy unchanged, and patient-mean coverage is unlikely to carry the residual breadth benefit. The missing information would then lie in properties that a patient mean does not express, and the next candidate would be a distance between the patch distributions of patients.

5 Results

The manipulation check passed in all three splits. The coverage ratio of the concentration contrast is $\rho_{\text{con}} = 0.63, 0.68, \text{ and } 0.58$, as anticipated in Table 2. Concentrating the cohort around one patient lowers patient-macro balanced accuracy by 8.30 percentage points (95% interval: 7.66 to 8.95), whereas coverage-maximizing selection raises it over random sampling by 1.04 points (0.44 to 1.66). The first interval lies above the one-point threshold and the second crosses it, so the outcome is the concentration-damage reading of Table 3.

5.1 Manipulation check

The split-level coverage distances match Table 2 to the third decimal. Averaged over splits, the clustered allocation covers validation patients at 0.352, the random allocation at 0.294, and the dispersed allocation at 0.269, against 0.386 for the five anchors. Mean pairwise distances within a cohort are 0.36, 0.59, and 0.67. Clustering thus removes about two thirds of the coverage that fifteen random additions contribute, and coverage-maximizing selection adds about a quarter of it. In 52 of the 90 class-split combinations, the class-level concentration ratio reaches 0.5: 19, 19, and 14 classes in the three splits.

The manipulation is uneven across cancer types (Table 7 in Appendix A). Class-level concentration ratios have a median of 0.58. Adrenocortical carcinoma, mesothelioma, pancreatic adenocarcinoma, uterine carcinosarcoma, and uveal melanoma stay below 0.25, and for uterine carcinosarcoma and uveal melanoma all three allocations cover validation patients equally well. At the other extreme, the clustered allocation of testicular germ cell tumours covers validation patients worse than the five anchors do, because the clustered cohort drops the other four anchors and the neighbours of the focal patient lie far from the validation patients those anchors cover. The selection ratio exceeds 0.5 only for breast invasive carcinoma and prostate adenocarcinoma. The focal patient’s site contributes 25 percent of the clustered patients and 17 percent of the random and dispersed patients; the two classes with few contributing sites, adrenocortical carcinoma and pheochromocytoma and paraganglioma, reach about 70 percent under every rule.

5.2 Concentration damage and selection gain

The clustered cohort is far less accurate than the other two (Table 4). Over all 30 classes, it reaches 62.99 percent, against 71.29 percent for the random and 72.33 percent for the dispersed cohort. It is also the most sensitive to which patients are drawn, with a standard deviation of 1.36 points across split-draw combinations against 0.66 points for both other cohorts.

Allocation	All classes		Site classes		Other classes	
	Accuracy [95% interval]	SD	Accuracy	SD	Accuracy	SD
Clustered	62.99 [61.68, 64.17]	1.36	57.61	2.53	67.10	2.06
Random	71.29 [69.93, 72.51]	0.66	69.11	1.24	72.96	1.56
Dispersed	72.33 [70.98, 73.57]	0.66	70.40	1.14	73.82	1.83

Table 4: At twenty patients with eight patches each, a clustered cohort is about eight points less accurate than a random one, and a dispersed cohort about one point more accurate. Accuracy is patient-macro balanced accuracy in percent, averaged over three splits and five draws, with test-patient 95% intervals. Site classes are the thirteen cancer types of Queisler (2026e); other classes are the remaining seventeen. SD is the population standard deviation across the fifteen split-draw combinations, in percentage points.

The concentration damage is $\Delta_{\text{con}} = 8.30$ points with a test-patient interval of 7.66 to 8.95 (Table 5). It is positive in all fifteen split-draw combinations, ranging from 5.23 to 10.12 points.

The interval lies entirely above the threshold, so a cohort concentrated around one patient trains a practically worse classifier than a random cohort of the same size, depth, and effective support.

The selection gain is $\Delta_{\text{sel}} = 1.04$ points (0.44 to 1.66). It is positive in all fifteen combinations, but ranges from 0.06 to 2.74 points. The interval excludes zero and crosses the threshold, so coverage-maximizing selection improves on random sampling without an established practically relevant gain. Section 4 rules out reading this interval as equivalence, because the underlying coverage change is only about a quarter of the coverage gained from five to twenty random patients.

Contrast	Classes	ρ	Estimate	95% interval	SD	Positive
Concentration damage Δ_{con}	All	0.63	8.30	[7.66, 8.95]	1.29	15 of 15
	Site	0.66	11.50	[10.77, 12.17]	2.07	15 of 15
	Other	0.60	5.86	[4.89, 6.84]	2.13	15 of 15
Selection gain Δ_{sel}	All	0.27	1.04	[0.44, 1.66]	0.71	15 of 15
	Site	0.38	1.29	[0.66, 1.96]	0.85	13 of 15
	Other	0.18	0.86	[-0.07, 1.75]	1.19	12 of 15

Table 5: The concentration damage lies above the one-point threshold in every class group, whereas every selection-gain interval crosses it. ρ is the coverage ratio of the contrast, ρ_{con} or ρ_{sel} , computed from coverage distances averaged over the group’s classes, splits, and draws. Estimates and paired test-patient intervals are in percentage points. SD and the positive count describe the contrast across the fifteen split–draw combinations.

The damage is larger than the descriptive scale of Section 3.1 anticipates. That scale, $\rho_{\text{con}} \times 5.2 \approx 3.3$ points, assumed that coverage carries the whole residual beyond effective support and that accuracy changes linearly with coverage distance. The observed damage is about two and a half times as large. It amounts to 87 percent of the 9.52-point advantage of twenty over five patients at the same budget (Queisler, 2026b). On the breadth–depth grid of that experiment, which uses other draws and is not paired with the present classifiers, five random patients with 32 patches reach 62.25 percent and twenty with eight patches 71.76 percent. Twenty patients concentrated around one patient are therefore about as accurate as five randomly drawn patients, whereas the random cohort of this experiment matches its grid counterpart within half a point. The selection gain, by contrast, stays below its linear scale of $0.27 \times 5.2 \approx 1.4$ points.

5.3 Where the damage arises

Both class groups show a concentration damage above the threshold, but it is twice as large in the site classes, at 11.50 points, as in the seventeen other classes, at 5.86 points (Table 5). The coverage ratios of the two groups differ much less, 0.66 against 0.60. The clustered cohort loses most in the site classes, at 57.61 percent, and the other classes include the five types that receive almost no concentration contrast. The selection gain follows the size of its coverage change: 1.29 points at a ratio of 0.38 for site classes and 0.86 points at 0.18 for the other classes, whose interval includes zero.

Within classes, the damage falls on test patients whom the clustered cohort covers worse (Table 6). In the tertile with the smallest coverage improvement, clustered training is 2.28 points more accurate than random training. In the middle tertile, the damage is 7.65 points, and in the tertile with the largest coverage improvement, 20.44 points, where recall under clustered training drops to 46.53 percent. The gradient appears in both class groups, most steeply in the site classes.

Site composition changes too little to account for most of this damage. Site signatures are visible in deep features of TCGA slides (Howard et al., 2021), so neighbours of one patient may share

Coverage-improvement tertile	All classes			Damage by class group	
	Clustered	Random	Damage	Site	Other
Low	74.57	72.29	-2.28	-1.33	-3.01
Middle	66.73	74.38	7.65	11.07	5.04
High	46.53	66.96	20.44	25.32	16.70

Table 6: The concentration damage grows with the coverage that the clustered cohort withholds from a test patient. Tertiles split each class’s test patients per draw by their coverage improvement, the distance to the nearest clustered minus the distance to the nearest random training patient. Recalls and damages are in percentage points, with the patient weights of the primary endpoint within a tertile and equal weight for each class, split, and draw. Estimates are descriptive and carry no intervals.

its site. The focal patient’s site contributes eight percentage points more of the clustered than of the random cohort, and eleven points more in the site classes, 19 against 8 percent. For these classes, doubling the number of sites at a fixed patient count changed accuracy by only 1.06 points (Queisler, 2026e), an order of magnitude less than their concentration damage.

6 Discussion and conclusion

At a fixed number of patients, patches, and effective support, which patients form a TCGA-UT training cohort matters. A cohort of twenty patients that resemble one patient in the Virchow2 embedding is 8.30 points less accurate than a random cohort, nearly the full advantage of twenty over five patients at the same budget, and this damage appears in every split-draw combination and in both class groups. Replacing random sampling by coverage-maximizing selection adds 1.04 points, a gain that is consistently positive but not established as practically relevant.

Two properties of the damage bear on its cause. First, it concentrates on test patients far from the clustered cohort, while patients near it are classified slightly better than under random training. This pattern is what missing coverage predicts, and a loss of independent observations alone would not be expected to raise recall for the patients the cohort covers well. Second, the damage is about two and a half times the linear coverage scale and nearly erases the breadth advantage. Accuracy may therefore respond non-linearly to coverage distance, with the clustered cohort crossing a region in which whole appearances of a cancer type go unrepresented. The limitations of Section 7 also apply in full: the clustered patients are redundant with one another beyond what effective support counts, and they over-represent one appearance, towards which the class boundary may shift. The tertile gradient favours coverage and appearance imbalance over redundancy alone but does not separate them, and the experiment does not show that patient-mean coverage carries the residual breadth benefit of random cohorts.

The asymmetry between the two contrasts matters for patient shortage. Random sampling at twenty patients already attains most of the coverage that coverage-maximizing selection can reach, and it attains most of the accuracy too. Cohort composition therefore protects mainly against a poorly covering cohort, such as one drawn from a single clinic’s typical cases, and offers at most a small gain when a random cohort of twenty patients is available. Under the concentration-damage reading, the value of coverage-guided selection remains open where random cohorts cover a class less well. The next step is the comparison of Section 4: coverage-selected against random cohorts of five patients with 32 patches each, the allocation in which patient shortage was first observed. An estimate of between-patient correlation within clustered and random cohorts would, in addition, show how much of the concentration damage effective support would attribute to redundancy once patient similarity is counted.

7 Limitations

The clustered allocation does more than withhold coverage. Its patients resemble each other, so they are more redundant with one another than effective support, which assumes independent patients, counts them; and they over-represent one appearance of the cancer type, towards which a classifier may shift its class boundary. A concentration damage therefore reflects between-patient redundancy, within-class imbalance of appearances, and missing coverage together, and the coverage-improvement tertiles separate these only partly.

The manipulation is uneven across cancer types. In small or homogeneous classes such as adrenocortical carcinoma, pancreatic adenocarcinoma, uterine carcinosarcoma, and uveal melanoma, the class-level ratio of the concentration contrast stays below 0.25 in most splits, so these classes contribute little contrast to the class average. Coverage is defined on patient means in one representation and a single cohort size, so a null result excludes coverage only as measured here. All results are conditional on frozen Virchow2 features, multinomial logistic regression, and the three overlapping splits, which are not independent replications.

References

- Frederick M. Howard, James Dolezal, Sara Kochanny, Jefree Schulte, Heather Chen, Lara Heij, Dezheng Huo, Rita Nanda, Olufunmilayo I. Olopade, Jakob N. Kather, Nicole Cipriani, Robert L. Grossman, and Alexander T. Pearson. The impact of site-specific digital histology signatures on deep learning model accuracy and bias. *Nature Communications*, 12:4423, 2021. doi: 10.1038/s41467-021-24698-1.
- Yannik Queisler. Patient shortage in histopathology: Do additional patients help without adding coverage? Technical report, TU Berlin, 2026a. Companion concentrated-coverage report.
- Yannik Queisler. Patient shortage in histopathology: Breadth, depth, and effective support. Technical report, TU Berlin, 2026b. Companion breadth–depth report.
- Yannik Queisler. Patient shortage in histopathology: Does full-feature redundancy explain the residual breadth benefit? Technical report, TU Berlin, 2026c. Companion full-feature redundancy report.
- Yannik Queisler. Patient shortage in histopathology: Is the residual breadth benefit a patient-coverage benefit? Technical report, TU Berlin, 2026d. Companion patient-coverage report.
- Yannik Queisler. Patient shortage in histopathology: Is the residual breadth benefit a site-coverage benefit? Technical report, TU Berlin, 2026e. Companion site-coverage report.
- Donald B. Rubin. The bayesian bootstrap. *The Annals of Statistics*, 9(1):130–134, 1981. doi: 10.1214/aos/1176345338.
- Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, George Shaikovski, Neil Tenenholtz, James Hall, Thomas Fuchs, Nicolo Fusi, Siqi Liu, and Kristen Severson. Virchow2: Scaling self-supervised mixed magnification models in pathology. *arXiv preprint arXiv:2408.00738*, 2024. doi: 10.48550/arXiv.2408.00738.

A Coverage by cancer type

Table 7 lists the manipulation check for each cancer type, averaged over the three splits and five draws.

Cancer type	Coverage distance					Focal-site share, %		
	r_5	r_{clu}	r_{ran}	r_{dis}	ρ_{con}	ρ_{sel}	clu / ran / dis	
Adrenocortical carcinoma	0.309	0.259	0.246	0.246	0.21	0.01	72 / 72 / 70	
Bladder urothelial carcinoma	0.573	0.555	0.430	0.394	0.87	0.25	12 / 6 / 6	
Brain lower grade glioma	0.261	0.228	0.196	0.167	0.49	0.45	22 / 8 / 6	
Breast invasive carcinoma	0.541	0.508	0.430	0.373	0.70	0.51	17 / 11 / 7	
Cervical squamous and endocervical carcinoma	0.521	0.491	0.383	0.344	0.78	0.28	11 / 9 / 8	
Colon adenocarcinoma	0.310	0.286	0.245	0.222	0.62	0.36	20 / 5 / 8	
Esophageal carcinoma	0.560	0.508	0.410	0.370	0.66	0.27	21 / 7 / 11	
Glioblastoma multiforme	0.324	0.281	0.245	0.212	0.45	0.42	43 / 14 / 14	
Head and neck squamous cell carcinoma	0.359	0.324	0.274	0.253	0.58	0.25	20 / 11 / 11	
Kidney chromophobe	0.203	0.167	0.153	0.141	0.28	0.23	16 / 15 / 18	
Kidney renal clear cell carcinoma	0.256	0.254	0.191	0.164	0.96	0.43	34 / 18 / 14	
Kidney renal papillary cell carcinoma	0.355	0.318	0.276	0.252	0.53	0.30	9 / 4 / 5	
Liver hepatocellular carcinoma	0.368	0.333	0.285	0.253	0.57	0.39	32 / 19 / 19	
Lung adenocarcinoma	0.500	0.469	0.385	0.341	0.73	0.38	20 / 8 / 7	
Lung squamous cell carcinoma	0.499	0.448	0.394	0.358	0.51	0.35	15 / 6 / 6	
Mesothelioma	0.512	0.421	0.398	0.391	0.21	0.06	7 / 5 / 6	
Ovarian serous cystadenocarcinoma	0.441	0.366	0.341	0.326	0.25	0.14	55 / 46 / 48	
Pancreatic adenocarcinoma	0.438	0.362	0.344	0.322	0.19	0.24	11 / 6 / 9	
Pheochromocytoma and paraganglioma	0.249	0.186	0.161	0.153	0.28	0.09	68 / 68 / 70	
Prostate adenocarcinoma	0.189	0.176	0.153	0.117	0.65	0.98	15 / 8 / 10	
Rectum adenocarcinoma	0.272	0.229	0.199	0.192	0.40	0.10	18 / 12 / 7	
Sarcoma	0.379	0.369	0.287	0.268	0.89	0.21	38 / 26 / 20	
Skin cutaneous melanoma	0.493	0.444	0.396	0.354	0.49	0.43	18 / 10 / 10	
Stomach adenocarcinoma	0.564	0.541	0.416	0.378	0.84	0.26	25 / 16 / 18	
Testicular germ cell tumours	0.274	0.418	0.198	0.181	2.87	0.21	47 / 39 / 46	
Thymoma	0.318	0.278	0.223	0.211	0.58	0.13	12 / 8 / 8	
Thyroid carcinoma	0.341	0.340	0.237	0.199	0.99	0.36	19 / 16 / 13	
Uterine carcinosarcoma	0.483	0.397	0.395	0.403	0.03	−0.09	14 / 12 / 11	
Uterine corpus endometrial carcinoma	0.439	0.411	0.339	0.309	0.72	0.31	15 / 4 / 7	
Uveal melanoma	0.239	0.183	0.182	0.183	0.02	−0.02	32 / 24 / 22	
Mean over classes	0.386	0.352	0.294	0.269	0.63	0.27	25 / 17 / 17	

Table 7: The concentration contrast varies widely between cancer types, and five types receive almost none of it. Coverage distances are averaged over the three splits and five draws, and the class-level ratios ρ_{con} and ρ_{sel} are computed from these averages. Focal-site shares give the percentage of the nineteen non-focal patients that come from the focal patient’s tissue source site in the clustered, random, and dispersed allocation.